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direction of motion. Find the direction in which A moves after collision.

**Ans:** $\theta = 26° 33'$

5. A bullet of mass 20 g strikes a pendulum of mass 5 kg. The centre of mass of pendulum rises a vertical distance of 10 cm. If the bullet gets embedded into the pendulum, calculate its initial speed.

**Ans:** $v = 351.4 ms^{-1}$

## V. Conceptual Questions

1. A spring which in initially in un-stretched condition, is first stretched by a length x and again by a further length x. The work done in the first case $W_1$ is one third of the work done in second case $W_2$. True or false?
2. Which is conserved in inelastic collision? Total energy (or) Kinetic energy?
3. Is there any net work done by external forces on a car moving with a constant speed along a straight road?
4. A car starts from rest and moves on a surface with uniform acceleration. Draw the graph of kinetic energy versus displacement. What information you can get from that graph?
5. A charged particle moves towards another charged particle. Under what conditions the total momentum and the total energy of the system conserved?

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